# 01 · PREPROCESSING & DATA PREPARATION PIPELINE

### From Raw Satellite Signals to Clean Deep Learning Tensors

**Project:** SAR-Guided Satellite Cloud Removal · ECRformer + Conditional PatchGAN
**Authors:** Agnal K V · Bivin Balu · Vaishnav T Prakash | **Supervisor:** Dr. Sabu M K
Department of Computer Applications, Cochin University of Science and Technology (CUSAT)

---

**Document Purpose & Core Philosophy**

This document breaks down the **data preprocessing pipeline** in the simplest, most intuitive manner. It details how multi-modal spaceborne observations (Sentinel-1 C-band Synthetic Aperture Radar and Sentinel-2 Multi-Spectral Optical imagery) are collected, radiometrically calibrated, masked for cloud probability, and structured into normalized 4D tensors before reaching the neural network. Crucially, it explains **why each engineering design choice was selected over prevailing alternatives**.

## 1 The End-to-End Preprocessing Flow

The preprocessing pipeline converts two fundamentally different physical phenomena—microwave radar backscatter and solar optical reflectance—into a unified, pixel-aligned tensor representation.

**Step-by-Step Data Flow Diagram:**

1. **Raw Data Acquisition:** Ingest SEN12MS-CR paired data: Sentinel-1 SAR (VV, VH bands) and Sentinel-2 L1C Optical Top-of-Atmosphere (TOA) scenes at $256 \times 256$ spatial resolution.
2. **Optical Reflectance Normalization:** Divide raw optical digital numbers by 10,000 to map physical solar reflectance to the bounded range $[0.0, 1.0]$. Select the standard 4-band subset: Red (B04), Green (B03), Blue (B02), and Near-Infrared (B08).
3. **SAR Radar Decibel Calibration:** Convert synthetic aperture radar power to decibels (dB), followed by physical sensor clipping:
   - VV channel clipped to $[-25.0, 0.0]$ dB $\xrightarrow{\text{linear min-max}}$ $[0.0, 1.0]$
   - VH channel clipped to $[-35.0, 0.0]$ dB $\xrightarrow{\text{linear min-max}}$ $[0.0, 1.0]$
4. **Offline Cloud Mask Generation (ESA `s2cloudless`):** Pass 13-band Sentinel-2 L1C spectra into a pre-trained LightGBM gradient-boosted decision tree using Cirrus (B10) and SWIR (B11/B12). Threshold raw cloud probability at $> 0.40$ to yield base binary mask $M_{\text{raw}}$.
5. **Online 3-Step Morphological Soft Mask Refinement:** On every dataloader fetch:
   - *De-speckle:* $9 \times 9$ Average Pooling (stride 1, padding 4) thresholded at $> 0.5$ (removes sensor noise).
   - *Dilation:* $5 \times 5$ Max Pooling (radius 2) to envelop semi-transparent optical haze and cloud halos.
   - *Gaussian Blur:* $11 \times 11$ spatial filter with $\sigma = 2.0$, producing continuous soft weights $M \in [0.0, 1.0]$.
6. **Shared Synchronized Spatial Cropping (Training Only):** Extract identical random $128 \times 128$ spatial bounding boxes across all 4 tensors $(S_{2,\text{cloudy}}, S_1, S_{2,\text{clear}}, M)$ to guarantee sub-pixel multi-modal registration.

# 2 Tensor Notation & Mathematical Transformations

In deep learning pipelines, spatial data is structured as 4D tensors: [Batch $B$, Channels $C$, Height $H$, Width $W$].

## 2.1 Data Transformation Summary Table

| Variable | Physical Origin | Raw Input | Processed Range | Tensor Shape (Train / Test |
|---|---|---|---|---|
| $S_{2,\text{cloudy}}$ | Sentinel-2 Optical (Cloudy) | $[0, 10000]$ DN | $[0.0, 1.0]$ Reflectance | $[B, 4, 128, 128] \rightarrow [B, 4, 256, 25$ |
| $S_1$ | Sentinel-1 SAR Radar | $[-\infty, +\infty]$ dB | $[0.0, 1.0]$ Normalized | $[B, 2, 128, 128] \rightarrow [B, 2, 256, 25$ |
| $M$ | ESA `s2cloudless` Soft Mask | Probability $[0, 1]$ | $[0.0, 1.0]$ Continuous | $[B, 1, 128, 128] \rightarrow [B, 1, 256, 25$ |
| $X_{\text{in}}$ | Generator Input Stack | Concatenation | $[0.0, 1.0]$ Multi-modal | $[B, 7, 128, 128] \rightarrow [B, 7, 256, 25$ |
| $S_{2,\text{clear}}$ | Cloud-Free Ground Truth | $[0, 10000]$ DN | $[0.0, 1.0]$ Reflectance | $[B, 4, 128, 128] \rightarrow [B, 4, 256, 25$ |

Table 1: Tensor shapes and ranges across preprocessing stages ($B = 4$ in training micro-batches, $B = 1$ in test).

## 2.2 Mathematical Formulations

**1. SAR Decibel Normalization ($S_1$):** Raw radar backscatter spans orders of magnitude. The calibration to physical reflectance bounds $[0, 1]$ is:

$$S_{1,\text{VV}} = \frac{\text{clamp}(S_{1,\text{raw_VV}}, -25.0, 0.0) + 25.0}{25.0} \tag{1}$$

$$S_{1,\text{VH}} = \frac{\text{clamp}(S_{1,\text{raw_VH}}, -35.0, 0.0) + 35.0}{35.0} \tag{2}$$

**2. Three-Step Morphological Mask Refinement ($M$):** Starting from the raw boolean mask $M_{\text{raw}} = \mathbb{I}(P_{\text{cloud}} > 0.40)$:

$$M_{\text{clean}} = \mathbb{I}\left(\text{AvgPool}_{9\times9}(M_{\text{raw}}) > 0.5\right) \quad \text{(Eliminates single-pixel noise)} \tag{3}$$

$$M_{\text{dilated}} = \text{MaxPool}_{5\times5}(M_{\text{clean}}) \quad \text{(Expands radius by 2 pixels over thin haze)} \tag{4}$$

$$M = \text{clamp}\left(G_{\sigma=2.0,k=11} * M_{\text{dilated}},\ 0.0,\ 1.0\right) \quad \text{(Produces smooth continuous soft boundary)} \tag{5}$$

# 3 Why We Used These Methods Over Others

Examiners frequently interrogate why specific preprocessing techniques were chosen instead of simpler alternatives. Below is the direct technical justification:

### 1. Why Multi-Modal SAR (Radar) + Optical instead of Optical-Only Interpolation?

**Alternative:** Optical temporal stitching or spatial inpainting (e.g., Bicubic interpolation, PatchMatch, or clear-sky temporal compositing).

- **Failure of Alternative:** Optical sensors cannot penetrate thick clouds. In monsoon seasons or tropical regions, persistent cloud cover lasts for weeks, rendering temporal optical compositing obsolete. Furthermore, inpainting without external ground structure hallucinates fictional buildings or roads.
- **Why Our Choice Wins:** Sentinel-1 C-band microwave radiation ($\lambda \approx 5.6$ cm) completely penetrates atmospheric water droplets and cloud formations unimpeded, 24/7 day and night. SAR provides real, unoccluded structural ground truth (surface roughness, building edges, water boundaries) to guide optical reconstruction.

### 2. Why ESA's `s2cloudless` instead of Simple RGB Color or Otsu Thresholding?

**Alternative:** RGB Brightness Thresholding ($R+G+B >$ threshold) or Otsu's automatic luminance binarization.

- **Failure of Alternative:** In visible optical RGB, clouds are bright white; however, snow-capped mountains, white concrete building roofs, salt flats, and coastal sand beaches are *also* bright white. Simple RGB thresholding misclassifies cities and coastlines as clouds, causing the model to needlessly replace pristine ground pixels.
- **Why Our Choice Wins:** `s2cloudless` utilizes Sentinel-2's invisible multi-spectral channels: **Band 10 (Cirrus, 1375 nm)** and **Bands 11 & 12 (Short-Wave Infrared / SWIR, 1610 & 2190 nm)**. Solar radiation in Band 10 is almost totally absorbed by atmospheric water vapor; hence, ground reflections are pitch black, while high-altitude cirrus clouds reflect intensely. This yields over 95% true positive cloud detection without false positives on urban buildings.

### 3. Why Logarithmic Decibel (dB) Scaling for SAR instead of Raw Linear Amplitude?

**Alternative:** Raw linear radar backscatter cross-section ($\sigma_0$).

- **Failure of Alternative:** Radar backscatter follows an extreme power-law distribution: flat calm water reflects nearly 0.0001, while a corner reflector on a metallic bridge reflects over 100,000.0. In raw linear scale, 99.9% of land terrain compresses into a tiny peak near zero, causing vanishing gradients and training divergence.
- **Why Our Choice Wins:** Decibel conversion ($\text{dB} = 10 \cdot \log_{10} \sigma_0$) maps this exponential distribution into a well-behaved linear bell curve spanning $[-35\text{ dB}, 0\text{ dB}]$. Normalizing this with sensor-specific clipping ($[-25, 0]$ for VV, $[-35, 0]$ for VH) guarantees well-conditioned gradients inside neural backpropagation.

### 4. Why a 3-Step Continuous Soft Mask instead of a Hard Binary 0/1 Mask?

**Alternative:** Standard binary step function $M \in \{0, 1\}$.

- **Failure of Alternative:** A hard binary edge produces severe, unnatural visual seams (checkerboard stitching artifacts) at cloud perimeters. Pixels right on the boundary jump abruptly from 100% real optical reflectance to 100% synthesized neural output.
- **Why Our Choice Wins:** Morphological dilation guarantees that faint, semi-transparent cloud fringes are enveloped. Gaussian blurring ($\sigma = 2.0$) creates a continuous gradient $M \in [0.0, 1.0]$. In testing, this enables linear alpha blending:

$$\hat{Y} = (1 - M) \odot S_{2,\text{cloudy}} + M \odot P_{\text{dir}} \tag{6}$$

ensuring mathematical continuity and zero boundary seam artifacts across the image.

### 5. Why Shared Synchronized Spatial Cropping instead of Bilinear Downsampling?

**Alternative:** Downsampling full $256 \times 256$ scenes to $128 \times 128$ using bilinear or bicubic interpolation.

- **Failure of Alternative:** Downsampling a satellite image by $2\times$ doubles the ground sampling distance (GSD) from 10 meters/pixel to 20 meters/pixel. This permanently destroys fine high-frequency features (two-lane roads, narrow canals, vehicle clusters, small buildings).
- **Why Our Choice Wins:** Random $128 \times 128$ sub-window cropping preserves the native 10 m/pixel physical resolution intact. Synchronizing the exact bounding box across all four tensors preserves rigorous pixel-level spatial co-registration.

## 4 15-Second Viva Examiner Defense Quick-Sheet

**Q: What is the input dimensionality to your neural network?**
*Answer:* The network accepts a 7-channel multi-modal tensor: 2 channels of Sentinel-1 SAR radar (VV and VH), 4 channels of Sentinel-2 cloudy optical (Red, Green, Blue, NIR), and 1 channel of continuous soft cloud mask $M$. Shape is $[B, 7, 128, 128]$ in training and $[1, 7, 256, 256]$ at inference.

**Q: Why did you pre-calculate the `s2cloudless` masks into `.npz` cache files?**
*Answer:* Running LightGBM decision trees on 13 spectral bands for thousands of scenes during training would stall the GPU data pipeline. Precomputing and caching into uncompressed `.npz` files accelerates training throughput by over 800%.

**Q: How does SAR "see" through clouds physically?**
*Answer:* Sentinel-1 transmits active C-band microwave radiation ($\approx 5.405$ GHz, $\lambda \approx 5.6$ cm). Rayleigh scattering from cloud water droplets ($\approx 10$ $\mu$m) is inversely proportional to $\lambda^4$. Because the radar wavelength is thousands of times larger than optical light, microwave radiation passes through clouds with essentially zero attenuation.